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Chair of Thermodynamics of Mobile Energy Conversion Systems (TME)
Prof. Marco Günther, RWTH Aachen University
Forckenbeckstrasse 4, 52074 Aachen

10 July 2026

Application: Research Assistant/Associate (PhD), Fuel Cell Systems (Ref. V000011113 SOFC / V000011114 PEM CFD & Water Transport / V000011115 PEM Aging)

Dear Prof. Günther,

I am applying for the Research Assistant/Associate PhD positions in fuel-cell systems at TME, across all three open topics (SOFC systems, PEM CFD and water transport, and PEM aging). At DLR I built the cathode-side control architecture for our aircraft fuel-cell system: a 0D lumped-parameter plant model of the cathode loop in MATLAB/Simulink, feed-forward, PI and decoupler control plus back-pressure-valve protection, and a Model Predictive Control approach on top, all now being tuned against real system measurement data. I am also building a complementary 1D lumped model in GT Suite, and I work in Dymola too, so Modelica is familiar ground alongside Simulink. That control work sits on top of the system engineering I did earlier at DLR: building the balance-of-plant sizing tools, selecting the cathode, thermal-management and sensory-system components, and designing the full system rig in CATIA V5.

This sits on two years of hands-on PEM fuel-cell experience across scales, from Germany's only student-built hydrogen fuel-cell vehicle at Ecogenium (1 kW) to a 100 kW, 1.5 t FCEV pickup truck at Robert Bosch to the 150-200 kW aircraft powertrain at DLR. At Ecogenium I am leading this season's fuel-cell system testing, integration and troubleshooting on our newly built water-cooled stack, building on the standalone test bench I engineered and commissioned there, where I ran purging, humidity-cycling and polarisation-curve campaigns. I correlate that test data against my own models the same way I did at Bosch, where I engineered a strategy that safeguards fuel-cell and HV-battery degradation and validated a Simulink operating-strategy model against measurement data, work I am now co-authoring for review at SAE. I also have a published paper on aircraft design configurations (IJSED) from earlier in my studies.

A PhD is genuinely half of what I am weighing for what comes next, alongside industry fuel-cell roles, and continuing at RWTH under Prof. Pischinger's chair, in the city where I already live and work, is an easy decision. My Master's thesis has not formally started, but its direction is confirmed as fuel-cell system control, building directly on the cathode-control work above. I would be glad to contribute to whichever of the three topics suits the group's priorities best: my control and modeling background speaks most directly to the SOFC systems position, and my test-bench and measurement-correlation experience to the CFD-validation and aging positions. The system-architecture and control principles I work with (balance-of-plant, control loops, MPC) carry over from PEM to SOFC, even though my hands-on stack experience so far is PEM. My English is fluent and my German is at A2-B1.

Yours sincerely,

Vallabh Pataneni